

Diploma in Data
Analysis

Introduction to Data Analysis

Lesson 8: Summary Notes



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Lesson Objectives

- Hypothesis testing: more small sample cases
- Estimation

Introduction

This lesson will wrap up small sample hypothesis testing by exploring the alternative left tailed and right-tailed cases, thereafter we will introduce estimation of parameters with will be our start to linear regression and creating models for datasets. Finally, we will wrap up the lesson by recapping the practical concepts of Module 1.



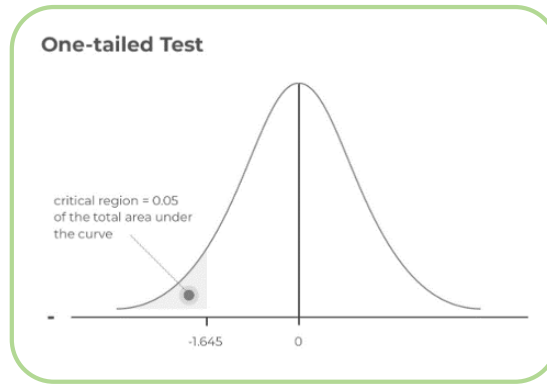
Small sample hypothesis

Recap

In the previous lesson, we explored the case of a small sample hypothesis test with an unknown population sample. A small sample indicates that the sample size is less than 30 observations. We use the student's t distribution because we can no longer assume that the distribution is normal.

Regions

In the previous lesson, we investigated the case of a two-sample hypothesis test for a small sample, which is the case where the distribution has rejection regions in both tails of the distribution. This lesson will cover one-tailed hypothesis testing.



Example

Let's understand this concept of a one-tailed test a bit better through the chocolate example.

Step 1

If the null hypothesis is that the mean weight for the population is 100g grams, the same as that of our sample, we can choose the alternative hypothesis to be smaller to that of a 100g.

$$H_0: \text{mean} \geq 100$$

$$H_a: \text{mean} < 100$$

To test this statement, select a total of 5 chocolate bars ($n = 5$) from the shop at an alpha () value of 0,05. We choose a 95% level of confidence in the hypothesis test.

This is a left-tailed test because we are dealing with a one-tailed test and we want to test the hypothesis that the mean weight is less than 100g.

Step 2

α .

Use the student's t-distribution table to calculate the critical t-value.

Degrees of freedom (v)	Amount of area in one tail (α)							
	0.0005	0.001	0.005	0.010	0.025	0.050	0.100	0.200
1	636.6192	318.3088	63.65674	31.82052	12.70620	6.313752	3.077684	1.376382
2	31.59905	22.32712	9.924843	6.964557	4.302653	2.919986	1.885618	1.060660
3	12.92398	10.21453	5.840909	4.540703	3.182446	2.353363	1.637744	0.978472
4	8.610302	7.173182	4.604095	3.746947	2.776445	2.131847	1.533206	0.940965
5	6.868827	5.893430	4.032143	3.364930	2.570582	2.015048	1.475884	0.919544
6	5.958816	5.207626	3.707428	3.142668	2.446912	1.943180	1.439756	0.905703
7	5.407883	4.785290	3.499483	2.997952	2.364624	1.894579	1.414924	0.896030
8	5.041305	4.500791	3.355387	2.896459	2.306004	1.859548	1.396815	0.888890
9	4.780913	4.296806	3.249836	2.821438	2.262157	1.833113	1.383029	0.883404
10	4.586894	4.143700	3.169273	2.763769	2.228139	1.812461	1.372184	0.879058
11	4.436979	4.024701	3.105807	2.718079	2.200985	1.795885	1.363430	0.875530
12	4.317791	3.929633	3.054540	2.680998	2.178813	1.782288	1.356217	0.872609
13	4.220832	3.851982	3.012276	2.650309	2.160369	1.770933	1.350171	0.870152
14	4.140454	3.787390	2.976843	2.624494	2.144787	1.761310	1.345030	0.868055
15	4.072765	3.732834	2.946713	2.602480	2.131450	1.753050	1.340606	0.866245
16	4.014996	3.686155	2.920782	2.583487	2.119905	1.745884	1.336757	0.864667

We have chosen the significance level for alpha to be 0,05 at a degree of freedom (d.f.) of 4. The sample size is 5, therefore the degree of freedom is:

$$d.f. = n - 1 = (5) - 1 = 4$$

The corresponding critical value in the table is 2,132. We are dealing with a left-tailed test; therefore, our rejection region is $(-2,132; -\infty)$.

Step 3

To calculate the test statistic, we use the formula and plug the values of the population mean, sample mean, sample standard deviation and sample size into the equation.

Assume that the population mean is 100g, the standard deviation is 1g, and the sample mean from the 5 chocolates is 99g.

The test statistic is -2,236.

$$\begin{aligned}\bar{x} &= 99 \text{ g}, \mu = 100 \text{ g}, s = 1, n = 5 \\ T &= \frac{\bar{x} - \mu}{s/\sqrt{n}} \\ T &= \frac{99 - 100}{1/\sqrt{5}} \\ T &= -2,236\end{aligned}$$

Step 4

We reject the null hypothesis if the test statistic is less than the critical value of -2,132.

Our t-value is less than -2,132, therefore the test statistic falls in the rejection region and we will reject the null hypothesis.

NOTES

Estimation

As data analysts, we often do not have a data set of the entire population we wish to explore. Therefore, we use sample statistics to infer or estimate the population parameters.

There are 2 types of estimates:

- **Point estimate**
 - The point estimate can be, for instance, the sample mean. A good estimate must be unbiased, consistent and relatively efficient.
 - For example, the expected sample mean has to be equal to the mean of the population, the value of the sample mean must approach the value of the population mean as the sample size increases and the sample mean has to contain the smallest possible variance.
- **Interval estimate**
 - An interval estimate can be a confidence interval that gives us an interval to estimate the range of values for the parameter with a certain level of confidence.
 - The level of confidence shows us the probability that the interval estimate will contain the parameter.

Least squares method

The least squares method is one of the most used methods in the estimation of parameters.

This method aims to find the line of best fit for a set of data by creating a straight line that minimises the sum of the squares of the errors that are generated by the results of the associated equations.

The least squares method draws a straight line through all the data points we have available in a scatter diagram for instance. It calculates the distance between each point and the line, which produces the residual value. The sum of squares, i.e the SSE, is the sum of each squared residual value.

$$\begin{aligned} \text{Residual} = e_i &= \text{observed} - \text{fitted value} \\ &= y - \hat{y} \end{aligned}$$

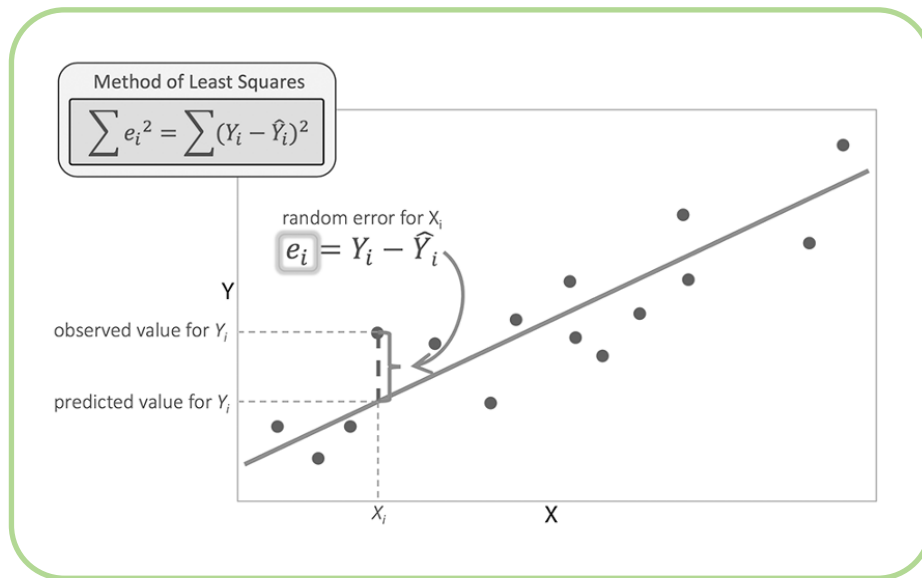
$$SSE = \sum e_i^2 = \sum_{i=1}^n (y - \hat{y})^2$$

Example

Suppose we measure the sound of the guitar in different temperatures.

We can plot the data points on a scatter plot with our corresponding decibel sound on the y-axis and the measured temperature at the time point on the x-axis.

Drawing a line of best fit through the data points we have plotted on the scatter plot will allow us to recognise what relationship there is between the speed of sound and temperature.



Together with linear regression, we can conclude that the speed of sound is greater in warmer temperatures and we will be able to hear sounds from far away more clearly.

NOTES

Resources

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